

GROSS ENGINEERING LLC

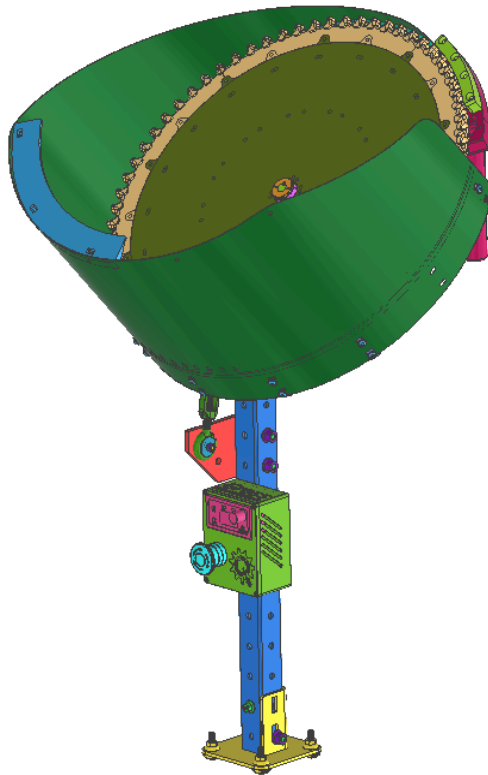
Design for Today & Innovate for Tomorrow

Commercial Case Orientation System

Customer Assembly & Operating Instructions

Pistol Configuration (.380 ACP – .45 ACP)

Revision A



Contents

1. Overview	3
2. What's in the Box	3
Main Bowl Assembly (pre-assembled)	3
Motor (ships separately)	3
Orientation Plate (ships separately)	3
Stand & Mounting	3
Power & Controls	3
Mounting Hardware & Fasteners	4
You Will Also Need (Customer-Supplied)	4
3. Before You Begin	4
Tools Required	4
Safety	4
4. Assembly	4
4.1 Unbox & Inspect	5
4.2 Assemble the Vertical Stand	5
4.3 Mount the Orientation Bowl	5
4.4 Install the Motor & Orientation Plate	5
5. Electrical Connection & First Power-On	6
Connections	6
Inductive Proximity Sensor Wiring	6
First Power-On	6
6. Using the Motor Controller	6
6.1 Running Interface — Controls	7
6.2 Entering Set Mode	7
6.3 Changing a Parameter Value	7
6.4 Adjustable Settings	7
7. Tuning & Setup	8
7.1 Finding Your Operating Speed	8
7.2 Installing & Trimming the Drop Hose	8
Sizing	8
7.3 Setting the Proximity Sensor Position	8
7.4 Setting the Bowl Angle (Turnbuckle)	9
8. Normal Operation	9
Loading the Bowl	9
Running	9
9. Troubleshooting	9
10. Maintenance	10
Cleaning the Bowl Shroud	10
Appendix A — Full Parameter Reference	11

1. Overview

The Commercial Case Orientation System is a 20-inch case orientation bowl that automatically orients and drops cases for downstream feeding. It uses a hybrid construction of commercial sheet-metal components and three custom parts that handle case orientation and drop.

The configuration you have received is set up for pistol use and handles cases from .380 ACP up to .45 ACP.

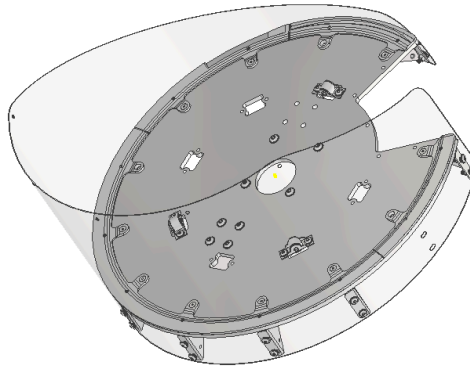
The system runs on 12V DC from the provided power supply and includes its own onboard controls: clutch engagement, speed control, and acceleration/deceleration control. Safety and stop functions include an emergency-stop (E-stop) button and an inductive proximity sensor that automatically stops the system when it detects a case in front of it.

2. What's in the Box

Your system ships partially assembled to protect the precision components in transit. When you unbox it, you will find:

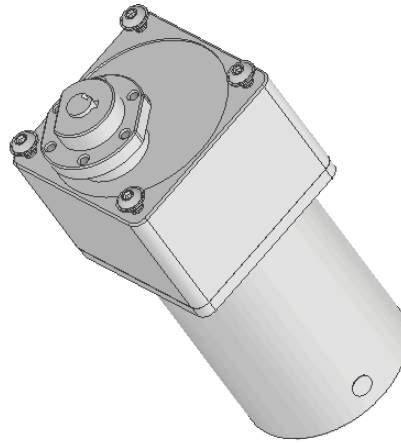
Main Bowl Assembly (pre-assembled)

- 20-inch orientation bowl with sheet-metal shroud affixed to the base plate
- Dropper assembly installed
- Wear plates installed
- Pivot / mounting-point hardware mounted to the bowl for shipping

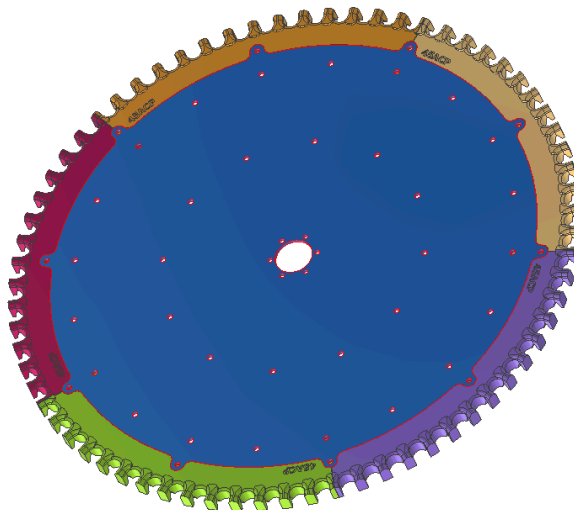


Motor (ships separately)

- Drive motor in its own foam mount, mounting hardware inserted, with a Wago-style connector on the power leads
- Drive coupler pre-fitted to the motor for shipping

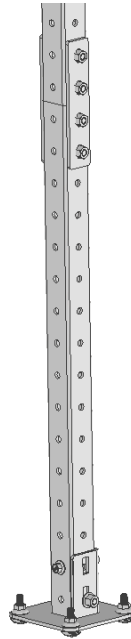
**Orientation Plate (ships separately)**

- Orientation plate with the pistol-specific inserts, packaged between the motor and the shroud



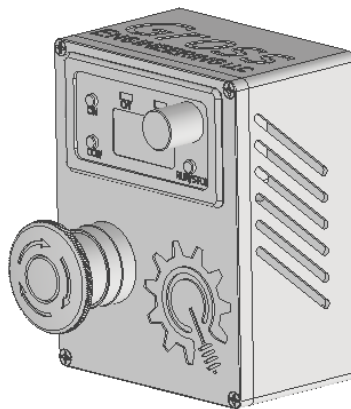
Stand & Mounting

- Two-piece vertical stand with joining gusset plates and mounting hardware
- Floor / bench mount plate with hardware to secure the stand to your floor or workbench



Power & Controls

- 120W power supply, 12V DC 10A (converts 110V AC → 12V DC)
- Modular power / control unit with integrated speed control and an E-stop button
- Remote-mount inductive proximity sensor with connectors



Mounting Hardware & Fasteners

The system is fully metric. Use the correct size at each location:

Fastener	Used For
M8 bolts with nyloc nuts	Large mounting fixtures, stand joints, and pivot points
M6 bolts	Motor attachment
M5 bolts	Bowl shroud
M4 bolts (3 provided)	Orientation plate (six holes; three for future use)
Turnbuckle	Bowl-angle adjustment, with its own lower pivot point

You Will Also Need (Customer-Supplied)

Feed / Drop Tube — A polyvinyl feed tube is required but not included. Recommended size: 3/4" outer diameter × 5/8" inner diameter. You will trim it to fit your application (see Section 7.2).

Floor / Bench Fasteners — Mounting fasteners for anchoring the stand are not provided. We recommend 3/8" lag bolts for a floor mount, or through-bolts for a benchtop mount (see Section 4.2).

3. Before You Begin

Tools Required

- 13mm wrench (and/or impact driver) — for the M8 stand and pivot hardware
- Hex / Allen keys to suit the M4, M5, and M6 socket hardware
- Drill and bits sized for your chosen floor / bench anchors
- A pencil or marker for tracing mounting-hole locations
- A small quantity of your brass for setup and tuning

Safety

- Keep the system unplugged while performing mechanical assembly.
- Be aware of where your hands are before applying power for the first time — the bowl may begin moving as soon as it is energized.
- The E-stop button instantly de-energizes the system. Familiarize yourself with it before running: press in to stop, twist clockwise to release.
- Anchor the stand securely before loading brass or running the system.

4. Assembly

4.1 Unbox & Inspect

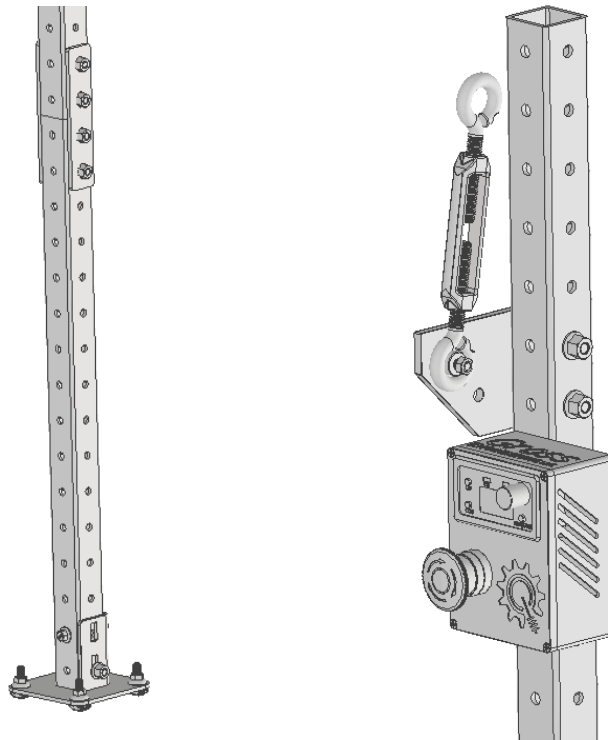
Unbox all items and separate them by group. Inspect every component for shipping damage before beginning assembly. If anything arrived damaged, stop and contact Gross Engineering before proceeding.

4.2 Assemble the Vertical Stand

We recommend fully assembling the vertical stand before mounting the bowl — it makes the rest of the process easier.

Mounting fasteners not provided — For securing the base, use 3/8" lag bolts for a floor mount, or through-bolts through the bench for a benchtop mount. Begin by tracing the mount's hole locations onto your chosen surface, then fasten the base accordingly.

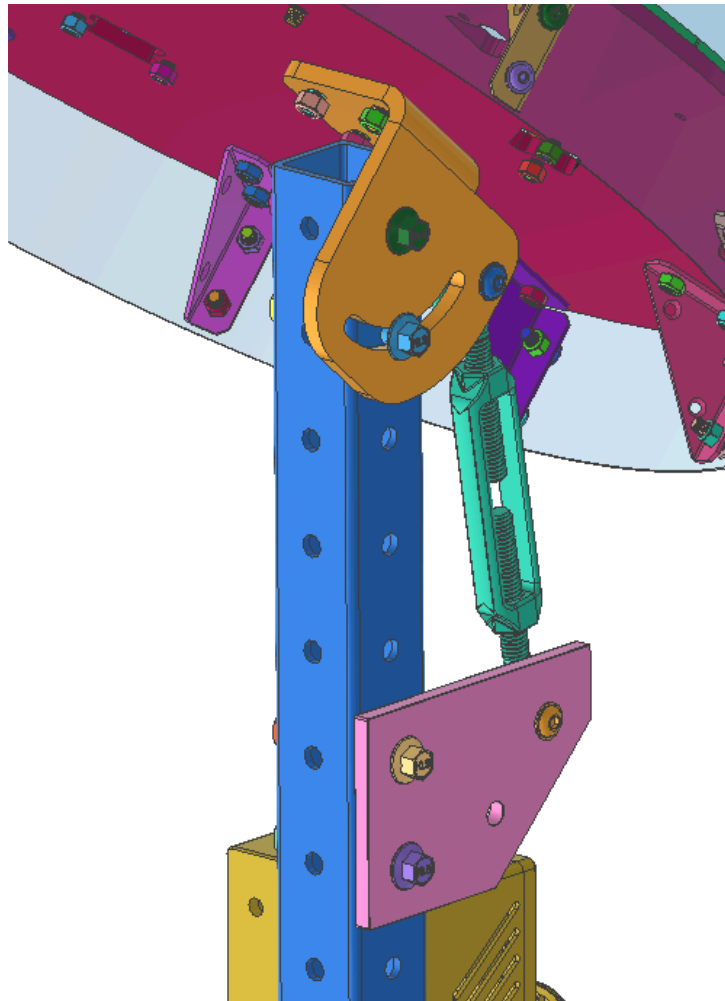
1. Take one of the two vertical members, line up the holes, and install the **2× M8 bolts with nyloc nuts**. Using a 13mm wrench and impact, tighten until firmly secured.
2. Install the **two black gusset plates** at the top end of the post. These are slotted so the two posts can sit on top of one another and tighten down into the best fit.
3. Take the next section and align it so the members rest on top of each other — this lets the load transmit through compression. Secure with **2× M8 bolts and nyloc nuts**.
4. Install the **lower gusset mount plate**. Exact height is give-or-take and left to your judgment — we recommend placing it near the bottom end of the top piece, but adjust to suit your setup.
5. Install the **lower eye of the turnbuckle**.



4.3 Mount the Orientation Bowl

With the stand assembled and anchored:

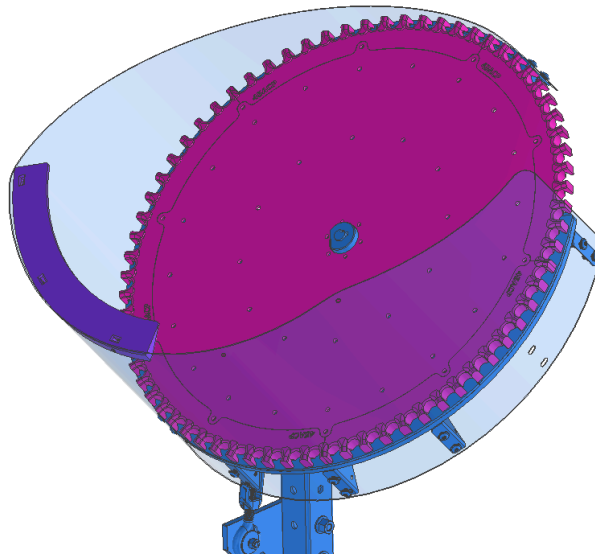
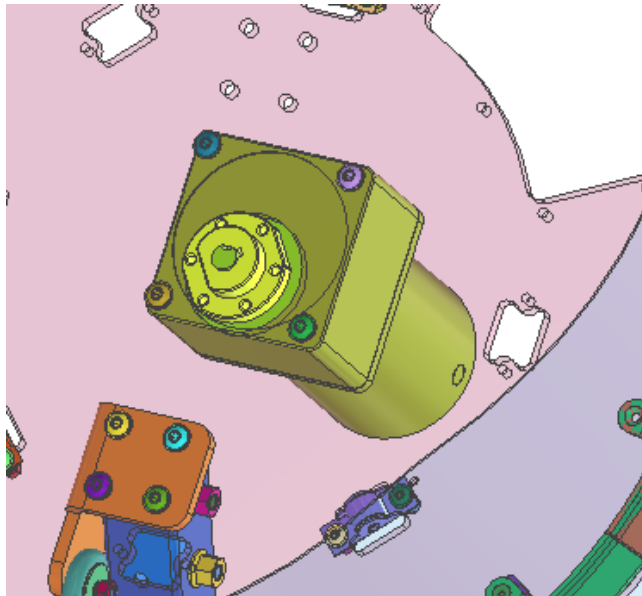
1. Carefully lift the orientation bowl and mount its pivot point to the **topmost hole** of the vertical member using an **M8 bolt and nyloc nut** — snug, but do not over-tighten.
2. Install the **second M8 nyloc** through the slotted arc of the pivot bracket on the underside of the case feeder — **leave this one loose** (it gets set during tuning).
3. Affix the **turnbuckle** to the pivot itself. The turnbuckle sets the bowl's working angle and is adjusted during tuning (Section 7.4).



4.4 Install the Motor & Orientation Plate

1. Install the provided motor into its location — it only fits one way. The **4× M6 bolts** hold it in place. Make sure it is properly secured before moving forward.
2. Before installing the orientation plate, check the **three bearings** that carry the majority of the rotational thrust load — confirm they are tight and snug.
3. Set the orientation plate on top of the coupler. You will see **six mounting holes**; install the **3× provided M4 bolts** to secure the plate to the bowl (the three extra holes are for future use).

Assembly complete — This concludes the physical assembly of the system. Proceed to the electrical connection.



5. Electrical Connection & First Power-On

The system is 12V DC.

Connections

1. The motor has **two output wires, red and black**. Match them to the red and black wires on the connector at the main control box — red to red, black to black.
2. Install the **two-prong connector** into its port on the underside of the control box.
3. Install the **three-prong connector** into its position on the control box.
4. Plug the **barrel-style power connector** into the box.

Rotation direction — For the bowl to work effectively it must turn clockwise. If it spins the wrong way on first power-up, flip the red and black motor wires to reverse the polarity and correct the rotation.

Inductive Proximity Sensor Wiring

The proximity sensor uses three wires. Match them to the corresponding wires on the connector:

Wire	Function
Brown	12V (load / power)
Blue	Ground
Black	Signal

The sensor has a red indicator light on its back end. With nothing in front of it, the light glows red; when a case is detected in front of it, the light turns off. You will use this indicator to set the sensor's trip position during tuning (Section 7.3).

First Power-On

Before plugging in for the first time, be aware of where your hands are. When you plug the system in, the main control screen should illuminate along with an LED indicator. The bowl may begin moving as well.

E-stop — To stop all motion, press the E-stop button in. This is a de-energizing switch — pressing it instantly removes all power from the bowl. To release, twist the button clockwise.

6. Using the Motor Controller



6.1 Running Interface — Controls

- **CW button** — Clockwise / forward rotation
- **CCW button** — Counter-clockwise / reverse rotation
- **Run / Stop button** — Start or stop rotation (also used to exit)
- **E-Stop button** — Immediately stops all motion; motion resumes once released (twist clockwise)
- **Potentiometer (rotary dial)** — Rotate to change rotational speed

6.2 Entering Set Mode

From the running interface, press and hold the potentiometer for 3 seconds to enter set mode. Once in set mode, rotate the potentiometer to move through the parameters. To save and exit, press and hold the potentiometer for more than 3 seconds.

6.3 Changing a Parameter Value

1. On the parameter you want, press the potentiometer to select the value for editing.
2. Rotate the potentiometer to change the value.
3. To edit by digit: press the potentiometer again to select a digit, then rotate to change that digit (0–9). Repeat for each digit.
4. Press the Run / Stop button to save the parameter.

6.4 Adjustable Settings

Most settings are factory-set — Your controller ships pre-configured. In normal use you will only adjust the small handful of settings below. Do not modify any parameter not listed here — changing locked parameters (especially F01) can cause the system to behave unpredictably. The complete parameter list is in Appendix A for reference only.

Code	What It Controls	How You'll Use It
F07	Forward speed (%)	Sets how fast the bowl runs in the forward direction.
F08	Reverse speed (%)	Sets how fast the bowl runs in reverse.

Code	What It Controls	How You'll Use It
F09	Overload current (clutch / jam stop)	The stall threshold that lets the system stop reliably on a jam. Leave as set; raise slightly if it nuisance-trips, lower if it isn't catching jams.
F11	Smooth stop / start	Set so the system eases to a stop and ramps smoothly back up to speed rather than starting and stopping abruptly.
F12	Acceleration / deceleration rate	Controls how quickly that ramp happens — 001 is the gentlest, 100 the most aggressive. Tune alongside F11 for the start/stop feel you want.

For everyday speed changes you can simply rotate the potentiometer on the running interface — there is no need to enter set mode.

7. Tuning & Setup

7.1 Finding Your Operating Speed

We recommend experimenting with different speeds to find what best suits your needs and your specific application.

In our testing, running the system at 30–40% speed reliably orients roughly 12,000–15,000 pieces. That range is a strong starting point, but every use case is different — start there and adjust up or down to suit your throughput, your cases, and how your downstream setup feeds.

7.2 Installing & Trimming the Drop Hose

The output drop hose (the 3/4" OD × 5/8" ID polyvinyl tube you supply) acts as a buffer between the orientation bowl and your filling application.

The bowl can drop cases faster than gravity carries them down the hose. If the sensor sits too close to the dropper, the system would cycle on and off constantly. By giving the hose some length, oriented cases stack up as a buffer above the prox sensor — typically 4–5 extra pieces of brass build up above the sensor before it trips. That buffer means the system isn't switching on and off every second: it builds a cushion, stops, then resumes as the buffer draws down.

Sizing

- Position the prox sensor **no closer than 6 inches** from the dropper — this is the minimum needed to develop the buffer.
- You can place it farther away for a larger buffer if you wish; 6 inches is the recommended minimum, not a target.
- Trim the bowl end to set this distance, then route the other end down to your filling application and trim to length there.

7.3 Setting the Proximity Sensor Position

The proximity sensor has an 8mm sensing window and is held by a grub screw that lets you slide it in and out to set its distance from the case path.

Different case diameters need different sensor spacing: the larger the case diameter, the closer the sensor should sit. Set the sensor to reliably catch the smallest case in your lineup, and the larger cases will follow.

1. Drop pieces of your brass through the sensor and watch the red indicator — it glows red with nothing present and turns off when a case is detected.
2. Loosen the grub screw and move the sensor inward or outward until cases trip it reliably as they pass.
3. Make sure brass still passes cleanly through without snagging.
4. Re-check with your specific caliber any time you change case size.

7.4 Setting the Bowl Angle (Turnbuckle)

The turnbuckle adjusts the angle of the bowl, which controls how cases orient and drop.

Before adjusting — Loosen the two bolts on the pivot and the lower angular mount at the top of the bowl mounts. They need to be loose so the bowl can pivot freely and you are not fighting the angle as you set it. (These are the two joints you left loose during assembly.)

Adjusting the angle:

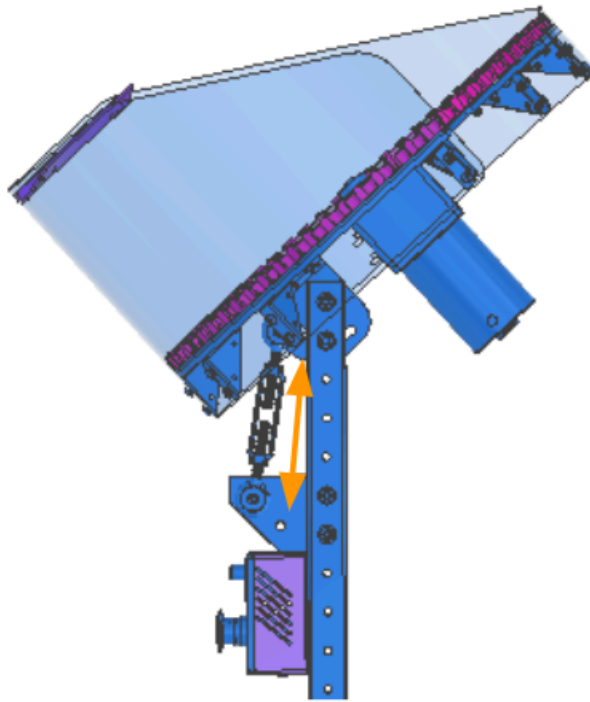
- **Lengthen** the turnbuckle → **increases** the bowl angle
- **Shorten** the turnbuckle → **decreases** the bowl angle

You are balancing two things as you adjust: cases that come around upside down should fall out at the top of the bowl's travel, while correctly oriented cases should stay in and continue. The angle also affects how brass rides up the lower side of the bowl as it cycles. Getting it right is a balancing act and takes some back-and-forth.

Procedure:

1. Put a handful of brass in the bowl and run it slowly — without feeding your application yet — so you can watch the orientation and drop behavior.
2. Adjust the turnbuckle until cases orient and reject/keep correctly.
3. Once it looks good, add a small amount more brass and confirm the angle holds.
4. When you are satisfied, take your **13mm wrench** and tighten both pivot points on the underside of the bowl. That locks in your set angle.

This angle may need occasional re-tuning over time — it is perfectly fine to adjust it as needed for different brass or conditions.



8. Normal Operation

Normal operation is simple: hook it up, run it, and tune the speed to your application.

Loading the Bowl

- Start with roughly 2,000–4,000 pieces of 9mm brass (or your case of choice).
- Fill the bowl until it is comfortably full — up to the point where it would overflow — then back off and remove brass as needed. That fill level becomes your working limit; load to it going forward.

Running

- Find a balance of speed that suits your application's demand (see Section 7.1).
- It is normal for the machine to run two to three cycles past what your application immediately needs — the buffer absorbs this, and the system will keep up with your application.
- If your application is outrunning the bowl (the bowl can't keep your fill supplied), that is your cue to **increase the system speed**.

9. Troubleshooting

Symptom	Likely Cause	What to Do
Bowl doesn't stop when brass is in front of the sensor	Prox sensor not working, not plugged in, or set too far out	Confirm the inductive prox is powered and plugged in. If it works but isn't reading, move it closer to the case path until it reliably detects brass.
Constant jams at the offload funnel	Drop wedge missing or incorrectly fitted	Try removing the drop wedge — or, if not installed, installing the drop wedge — to clear the offload path.
Brass dropping sideways, binding, or getting pulled over	Speed too high for a clean drop	Slow the system down to give cases a clean, controlled drop. If slowing down doesn't resolve it, contact Gross Engineering and we'll work out a solution for your application.
Brass catching underneath the orientation plate	Plate sitting too high	Lower and reseal the orientation plate.
Brass falling out the sides at the bottom of the bowl	Too much brass loaded, or bowl angle needs tuning	Reduce the fill level and/or re-tune the bowl angle (Section 7.4).

When to contact us — If a symptom persists after the steps above, or your application has needs outside this guide, contact Gross Engineering and we will work a solution for your specific application.

10. Maintenance

Cleaning the Bowl Shroud

The outer bowl shroud can be removed for cleaning. In normal operation there should be a maintained air gap between the shroud and the main mount points — this gap allows debris to evacuate while running. If that gap isn't set up correctly, or debris is building up, you can adjust the shroud spacing or remove the shroud / bowl to clean out any accumulation.

Periodic checks — The bowl angle and the pivot hardware may need occasional re-tightening or re-tuning over time, especially after changing case sizes. It is normal and expected to adjust these as part of routine upkeep.

Appendix A — Full Parameter Reference

For reference only — The table below lists every controller parameter. Only the highlighted user-adjustable parameters (F07, F08, F09, F11, F12) should be changed in normal use. All others are factory-set — do not modify.

Code	Function	Notes
F01	Working mode selection	Do not modify
F02	Delay A	Cycle timing (factory-set)
F03	Delay B — forward delay	Cycle timing (factory-set)
F04	Delay C — reverse delay	Cycle timing (factory-set)
F05	Delay D — stop-after-reverse delay	Cycle timing (factory-set)
F06	Work cycle count	0 = infinite loop (factory-set)
F07	Forward speed (% of unit)	User-adjustable
F08	Reverse speed (% of unit)	User-adjustable
F09	Overload current limit (clutch / stall)	User-adjustable — leave as set unless mitigating stalls/jams
F11	Motor stop behavior	User-adjustable. 0 = decelerate to stop & ramp back up; 1 = immediate stop
F12	Acceleration / deceleration level	User-adjustable. 001 = slowest, 100 = fastest
F13	LCD backlight brightness	Display preference

Set-mode reminder — Press and hold the potentiometer 3 seconds to enter set mode; rotate to select a parameter; press to edit, rotate to change; press Run/Stop to save; press and hold 3+ seconds to save and exit.